

Unit 1 Ext. Problems

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MATH 339 — Gonzaga University

compiled with L^AT_EX

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Note: sorry this was late, but i remember you saying it would be okay if i got these to you by monday, since you wouldnt grade them by monday. apologies

Problem 1A: §1.8 #12

write-up

this is a "Leontief input-output (open) model". each sector's production X has to cover what the other sectors use (AX) plus the outside demand (D). we had not covered this model in class, so I loosely learned it from:

- "Solving a Leontief Input-Output Model using Matrix Algebra" (youtube)
<https://www.youtube.com/watch?v=ucrZqMww1Qc>

the system is 10×10 with decimals (we do NOT want to do this by hand), so you can solve it with python (NumPy)

work

$$X = AX + D \Rightarrow X - AX = D \Rightarrow (I - A)X = D$$

$I - A$: diagonal entries $1 - a_{ii}$, non diagonal entries $-a_{ij}$.

big row reduce and solve with numpy $[I - A \mid D]$:

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \\ x_8 \\ x_9 \\ x_{10} \end{bmatrix} \approx \begin{bmatrix} 56.39 \\ 17.47 \\ 23.83 \\ 26.10 \\ 57.07 \\ 41.46 \\ 53.25 \\ 30.57 \\ 43.02 \\ 54.95 \end{bmatrix}$$

check: $X - AX = D$. ✓

$$X \approx (56.39, 17.47, 23.83, 26.10, 57.07, 41.46, 53.25, 30.57, 43.02, 54.95)$$

Problem 1B

write-up

this is a linear difference equation with a migration matrix M . each day, the cars move between locations by $\mathbf{x}_{k+1} = M\mathbf{x}_k$. we didn't cover this in class, so i loosely learned it from parts of the video:

- “Population Migration Models” (youtube)
<https://www.youtube.com/watch?v=HraKEoXkaho>
- “Linear difference equation with matrices” (math stack exchange)
<https://math.stackexchange.com/questions/2183669/>

mon to wed is 2 days, so 2 steps. then we do the multiplication with python (NumPy again).

work

$$M = \begin{bmatrix} 0.97 & 0.05 & 0.10 \\ 0.00 & 0.90 & 0.05 \\ 0.03 & 0.05 & 0.85 \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} 295 \\ 55 \\ 150 \end{bmatrix} \begin{array}{l} \text{Airport} \\ \text{North} \\ \text{Valley} \end{array}$$

tuesday:

$$\mathbf{x}_1 = M\mathbf{x}_0 = \begin{bmatrix} 0.97(295) + 0.05(55) + 0.10(150) \\ 0(295) + 0.90(55) + 0.05(150) \\ 0.03(295) + 0.05(55) + 0.85(150) \end{bmatrix} = \begin{bmatrix} 286.15 + 2.75 + 15 \\ 0 + 49.5 + 7.5 \\ 8.85 + 2.75 + 127.5 \end{bmatrix} = \begin{bmatrix} 303.9 \\ 57.0 \\ 139.1 \end{bmatrix}$$

wednesday:

$$\mathbf{x}_2 = M\mathbf{x}_1 = \begin{bmatrix} 0.97(303.9) + 0.05(57) + 0.10(139.1) \\ 0(303.9) + 0.90(57) + 0.05(139.1) \\ 0.03(303.9) + 0.05(57) + 0.85(139.1) \end{bmatrix} = \begin{bmatrix} 294.783 + 2.85 + 13.91 \\ 0 + 51.3 + 6.955 \\ 9.117 + 2.85 + 118.235 \end{bmatrix} = \begin{bmatrix} 311.543 \\ 58.255 \\ 130.202 \end{bmatrix}$$

check: $311.543 + 58.255 + 130.202 = 500$. ✓

wednesday $\approx$ 312 airport, 58 north, 130 valley

Problem 1C

write-up

D is a flexibility matrix: it turns forces $\mathbf{f}$ into deflections $\mathbf{y}$ by $\mathbf{y} = D\mathbf{f}$. the stiffness matrix D^{-1} goes the other way: $\mathbf{f} = D^{-1}\mathbf{y}$. we didn't cover this in class, so i loosely learned it from:

- “Relationship between Stiffness and Flexibility Matrix Derivation” (youtube)
<https://www.youtube.com/watch?v=2Z1Yp0w1lss>
- and some ai tools since i was very confused on this one.

work

i.

$$\mathbf{y} = D\mathbf{f} = \begin{bmatrix} 0.011 & 0.003 & 0.001 \\ 0.003 & 0.009 & 0.003 \\ 0.001 & 0.003 & 0.011 \end{bmatrix} \begin{bmatrix} 40 \\ 50 \\ 30 \end{bmatrix}$$

$$= \begin{bmatrix} 0.011(40) + 0.003(50) + 0.001(30) \\ 0.003(40) + 0.009(50) + 0.003(30) \\ 0.001(40) + 0.003(50) + 0.011(30) \end{bmatrix} = \begin{bmatrix} 0.44 + 0.15 + 0.03 \\ 0.12 + 0.45 + 0.09 \\ 0.04 + 0.15 + 0.33 \end{bmatrix} = \begin{bmatrix} 0.62 \\ 0.66 \\ 0.52 \end{bmatrix}$$

$$\boxed{y_1 = 0.62, y_2 = 0.66, y_3 = 0.52 \text{ inches}}$$

ii.

$$D = \frac{1}{1000}B, \quad B = \begin{bmatrix} 11 & 3 & 1 \\ 3 & 9 & 3 \\ 1 & 3 & 11 \end{bmatrix} \quad \Rightarrow \quad D^{-1} = 1000 B^{-1}$$

$$\det B = 11(99 - 9) - 3(33 - 3) + 1(9 - 9) = 990 - 90 + 0 = 900$$

$$\text{adj } B = \begin{bmatrix} 90 & -30 & 0 \\ -30 & 120 & -30 \\ 0 & -30 & 90 \end{bmatrix}$$

$$D^{-1} = \frac{1000}{900} \text{adj } B = \begin{bmatrix} 100 & -\frac{100}{3} & 0 \\ -\frac{100}{3} & \frac{400}{3} & -\frac{100}{3} \\ 0 & -\frac{100}{3} & 100 \end{bmatrix}$$

check: $DD^{-1} = I$. ✓

$$\mathbf{f} = D^{-1}\mathbf{y} = \begin{bmatrix} 100 & -\frac{100}{3} & 0 \\ -\frac{100}{3} & \frac{400}{3} & -\frac{100}{3} \\ 0 & -\frac{100}{3} & 100 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0.04 \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{100}{3}(0.04) \\ 100(0.04) \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{4}{3} \\ 4 \end{bmatrix}$$

$$\boxed{f_1 = 0, f_2 = -\frac{4}{3} \approx -1.33, f_3 = 4 \text{ lbs}}$$

negative f_2 means that force pushes up?

Problem 2B

write-up

i picked ii (SVD) and iii (spectral factorization). both use the fact that a square matrix with $Q^T Q = I$ has $Q^{-1} = Q^T$, and that powers of a diagonal matrix are just powers of its diagonal entries. i loosely learned the matrix powers part from:

- “How to Compute Matrix Powers Using Diagonalization” (YouTube)
<https://www.youtube.com/watch?v=xznLepig39g>
- and some ai tools since i was very confused on this one.

ii. singular value decomposition

U and V are $n \times n$ with $U^T U = I$ and $V^T V = I$, so

$$U^{-1} = U^T, \quad V^{-1} = V^T, \quad (V^T)^{-1} = V$$

D is diagonal with $\sigma_i > 0$, so

$$D^{-1} = \begin{bmatrix} 1/\sigma_1 & & \\ & \ddots & \\ & & 1/\sigma_n \end{bmatrix}$$

$A = UDV^T$ is a product of invertible matrices, so A is invertible:

$$A^{-1} = (V^T)^{-1}D^{-1}U^{-1}$$

$$\boxed{A^{-1} = VD^{-1}U^T}$$

check:

$$AA^{-1} = UDV^T VD^{-1}U^T = UDI D^{-1}U^T = UIU^T = UU^T = I \checkmark$$

iii. Spectral Factorization

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD(P^{-1}P)DP^{-1} = PD^2P^{-1}$$

$$A^3 = A^2A = (PD^2P^{-1})(PDP^{-1}) = PD^2(P^{-1}P)DP^{-1} = PD^3P^{-1}$$

the middle $P^{-1}P = I$ cancels every time, so

$$A^k = PD^kP^{-1}$$

D is diagonal, so just raise each entry to the power:

$$D^2 = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad D^3 = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad D^k = \begin{bmatrix} 2^k & 0 & 0 \\ 0 & 3^k & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\boxed{A^2 = P \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix} P^{-1}, \quad A^3 = P \begin{bmatrix} 8 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 1 \end{bmatrix} P^{-1}, \quad A^k = P \begin{bmatrix} 2^k & 0 & 0 \\ 0 & 3^k & 0 \\ 0 & 0 & 1 \end{bmatrix} P^{-1}}$$